



SCHOOL OF ADVANCED TECHNOLOGY

SAT301 FINAL YEAR PROJECT

*Enhancing Outdoor Running Experience
with a Mixed Reality Virtual Companion*

Final Thesis

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science

Abstract

Outdoor running is shaped not only by distance and pace, but also by runners' emotions and sense of engagement. Existing running apps rely on static numeric dashboards with limited emotional support. This study explores a Mixed Reality companion system for outdoor running on a Meta Quest 3 headset, aiming to enhance companionship and motivation through an embodied and interactive running experience.

Following two pre-study surveys ($N \approx 100$ each) on preferred running metrics and companion appearance, a virtual cartoon dog companion was designed and implemented with real-time physiological feedback. A controlled within-subjects A/B study ($N=12$) was conducted during outdoor running, comparing a companion mode against a panel-only mode. Results showed significantly higher companionship and motivation scores in the companion condition (p-value less than 0.001), with no significant difference in physiological measures or perceived workload.

The findings suggest that mixed reality virtual companions can enhance emotional and motivational aspects of outdoor running without increasing physical or cognitive burden beyond traditional fitness data visualization.

Keywords: Mixed Reality; User Experience; Running

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List of Acronyms

Term	Initial components of the term
3D	Three-Dimensional
AI	Artificial Intelligence
AR	Augmented Reality
BLE	Bluetooth Low Energy
CSV	Comma-Separated Values
HMD	Head-Mounted Display
HUD	Heads-Up Display
MR	Mixed Reality
NASA-TLX	NASA Task Load Index
SSQ	Simulator Sickness Questionnaire
VR	Virtual Reality

Chapter 1

Introduction

1.1 Motivation, Aims and Objective

Outdoor running is a common form of exercise, yet running alone is often perceived as monotonous and lacking in social engagement. While existing running applications such as Strava and Apple Fitness provide real-time performance metrics, they offer limited support for the emotional and social aspects of the running experience. Mixed Reality (MR) technology presents a promising opportunity to bridge this gap by embedding virtual content directly into the runner's physical environment. This project therefore explores the design and evaluation of an MR virtual companion for outdoor running, with the aim of enhancing companionship and motivation.

This study pursues four objectives: (1) conducting pre-study surveys to inform design decisions; (2) implementing a real-time MR companion system on a Meta Quest 3 headset; (3) evaluating the system through a controlled outdoor user study; (4) analyzing quantitative and qualitative data, including both subjective and objective measures, to assess the impact of the companion.

The project is guided by the following research question: Does an MR virtual companion improve runners' sense of companionship and motivation compared to a panel-only mode during outdoor running?

1.2 Literature Review

Social Support in Running: Given that running is frequently performed in isolation, researchers have increasingly focused on how social connection can transform the exercise experience. Notably, Hewer et al. [1] examined social reward and support among parkrun participants, finding that social connection during exercise was associated with greater enjoyment, reduced perceived fatigue, and higher performance outputs, suggesting that the

presence of others provides both psychological and physiological benefits during running. Franken et al. [2] similarly found that recreational runners who trained with partners were more likely to maintain consistent training regimens over time, highlighting the role of companionship in long-term exercise adherence. Beyond performance, Rapp and Tirabeni [3] investigated how amateur athletes experience self-tracking tools during sport, finding that while these tools effectively support data monitoring, they have limited impact on the social and emotional dimensions of the exercise experience. Together, these findings suggest that companionship and social support are meaningful components of running that existing data-centric tools do not address.

Running Technology and Its Limitations: The limitations of conventional running applications have motivated growing interest in more immersive approaches to running support. Li et al. [4] conducted a comprehensive survey of immersive running visualization research, mapping the current landscape and identifying the shift from static numeric dashboards toward richer, contextually embedded data experiences. Building on this, Li et al. [5] explored embedded and situated visualization in MR to support interval running, demonstrating that MR-based data overlays are technically feasible in outdoor running contexts. Beyond visualization, Armandi et al. [6] presented an AR system for outdoor jogging that provides real-time workout adaptation through a virtual coaching avatar, while Siriaraya et al. [7] investigated spatialized audio AR to allow runners to compete against their own past records, finding significant improvements in running performance. While these systems demonstrate the potential of AR and MR to enrich the running experience, they remain focused on performance optimization and data presentation rather than emotional engagement or companionship.

Virtual Companions in Exercise: A parallel body of research has explored the role of virtual companions in supporting motivation and emotional engagement in health and exercise contexts. Bickmore and Picard [8] demonstrated that relational agents capable of building and maintaining long-term relationships with users significantly enhanced engagement and trust, establishing a theoretical foundation for virtual companions in health contexts. Research on simulated exercising peers has extended this to physical activity: Silacci et al. [9] investigated the design of virtual agents as exercise companions rather

than coaches, exploring how they can provide social support and sustain motivation. A subsequent longitudinal study by Silacci et al. [10] directly compared exercising alone, with a human peer, and with an AI-driven virtual peer, finding that while human peers evoked stronger social presence, AI peers provided steadier and more consistent encouragement over time. Feng et al. [11] further found that companion-type virtual agents fostered stronger emotional engagement and parasocial relationships with users than task-oriented agents, suggesting that the design orientation of a virtual agent can meaningfully shape user experience. Collectively, these studies indicate that virtual companions hold genuine potential for enhancing the emotional dimensions of exercise, yet this potential remains unexplored in the context of outdoor MR running.

1.3 Research Gap

To the best of the author's knowledge, no prior work has specifically examined how a virtual companion embedded in an MR environment affects runners' sense of companionship, motivation, and overall experience during real outdoor running. This project addresses this gap by designing, implementing, and evaluating an MR virtual companion system in an authentic outdoor running setting.

1.4 Thesis Structure

The remainder of this thesis is structured as follows. Chapter 2 describes the methodology, covering the pre-study surveys, the system implementation, and the user study design. Chapter 3 presents the corresponding results from these three stages. Chapter 4 discusses the findings, acknowledges limitations, and outlines directions for future work. Finally, Chapter 5 concludes the report.

Chapter 2

Methodology

This project followed a three-phase research design: pre-study surveys to inform design decisions (Section 2.1), system implementation (Section 2.2), and a controlled user study to evaluate the system (Section 2.3).

2.1 Pre-Study Surveys

To ensure that design decisions were grounded in actual user preferences rather than assumptions, two independent online surveys were conducted prior to system development. The two surveys were distributed through separate channels via social media and messaging platforms, with instructions that participants should not complete both, in order to maintain sample independence. Full survey instruments are provided in Appendix A.

2.1.1 Running Metrics Survey

This survey aimed to identify which running metrics runners consider most important during a run. To ensure the responses reflected the experience of active runners, only participants who reported a regular running habit proceeded to the main questions. Participants were asked to select all metrics they would want to see while running from a list of eight commonly used running metrics: distance covered, time elapsed, current speed, average speed, heart rate, calories burned, route, and altitude change, as well as any additional metrics of their own choice. For participants who selected current speed or average speed, an additional question asked whether they preferred pace (min/km) or speed (km/h) as the display unit.

2.1.2 Companion Appearance Survey

This survey aimed to identify the most preferred visual design for the virtual running companion. Rather than asking participants to evaluate a small set of pre-selected designs, a systematic three-dimensional design space was constructed to ensure broad coverage: Kind (biological vs. robotic), Locomotion (biped, quadruped, jump, fly, wheel), and Realism (cartoon vs. realistic). 18 distinct candidate images were generated, covering the full combination of design dimensions. In the questionnaire, images were presented in a randomized order unique to each participant and without labels, to avoid priming effects. Participants were asked to select up to three preferred designs and optionally provide a free-text reason for their choices.

2.2 System Implementation

2.2.1 Hardware & Software

The system was developed in Unity (C#) and deployed on a Meta Quest 3 headset, operating in MR passthrough mode. Meta Quest 3 was selected for its full-color passthrough capability, stable development environment, and comprehensive SDK support. Real-time heart rate data was streamed via Bluetooth Low Energy (BLE) from a wearable device, Huawei Band 8, selected for its native BLE broadcast capability, allowing direct data reception within Unity through a standard BLE communication workflow. The companion character was created using publicly available 3D model and animation assets and further assembled and animated in Blender before being integrated into the Unity scene. The full project repository is available in Appendix B.

2.2.2 System architecture

As illustrated in Figure 2.1, the system is structured into three functional layers. In the input layer, real-time heart rate data is received from the Huawei Band 8 via BLE, and HMD displacement data is obtained from the built-in tracking system of Meta Quest 3, with system time serving as an additional reference for time-based calculations. These inputs are fed into the running metrics calculation module, which computes distance, pace, and

calories in real time. The computed metrics are then distributed through three paths: they are streamed directly to the HUD display for continuous visualization, passed to the trigger module for event generation based on predefined thresholds, and routed to the data recording module for session logging. Upon completion of a run, the recorded data is used to generate a results summary panel and exported as CSV files.

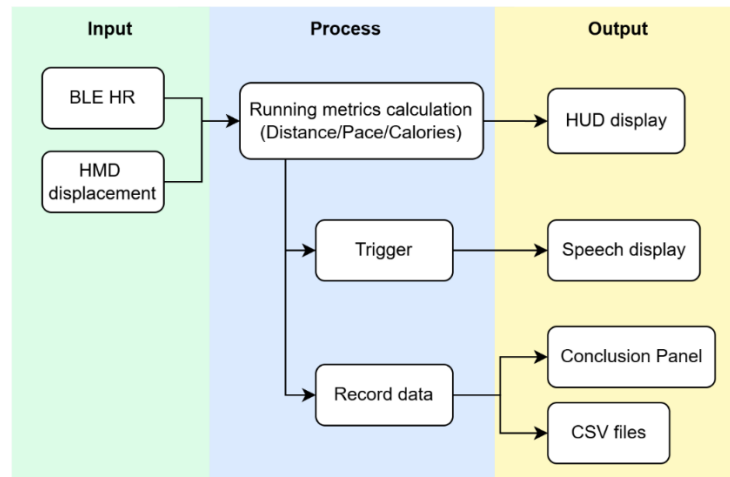


Figure 2.1 System architecture overview

As illustrated in Figure 2.2, the system workflow spans three distinct scene stages: Startup, Menu, and Test. During Startup, the system loads the cross-scene data manager and heart rate module for global initialization. In the Menu stage, the system continuously checks whether the BLE heart rate device is connected; if not, it remains in a waiting state while displaying a connection log. Once connected, the researcher selects a participant ID and test mode before proceeding to the Test stage. Tapping the Start button simultaneously initializes the timer, running metrics calculation, and data recording. Throughout the session, the HUD updates in real time while the trigger system monitors metrics and the dialogue system responds to threshold events. When the three-minute timer elapses, recording stops, the CSV is exported, and the results panel is displayed. The participant then returns to the menu, where the system checks whether both modes have been completed; if not, the session repeats for the remaining mode, and the workflow terminates only when both modes are finished.

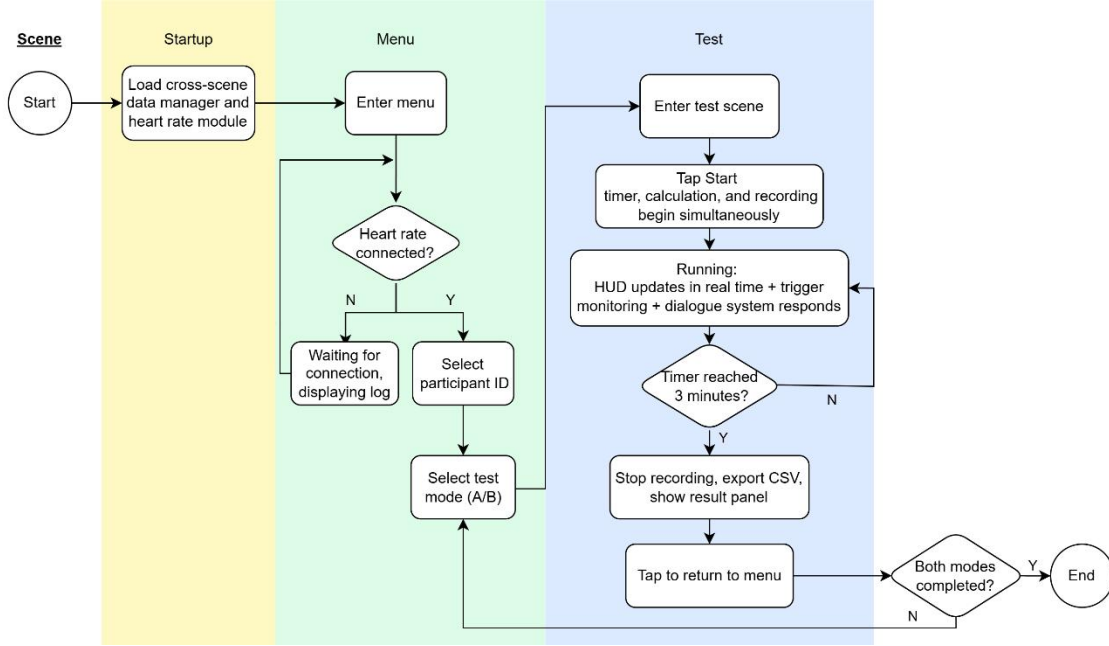


Figure 2.2 System operation flowchart

2.3 User study

2.3.1 Experimental Design

A controlled within-subjects A/B study was adopted, in which every participant experienced both experimental conditions, allowing direct intra-participant comparison and reducing the influence of individual differences. Condition order was counterbalanced across participants to control for learning and fatigue effects. Mode A (panel-only, control condition) presented running metrics on a HUD panel with mechanical-toned text prompts at predefined trigger points. Mode B (companion, experimental condition) used identical HUD layout and trigger logic, but delivered feedback through the 3D cartoon dog companion via animated reactions and conversational speech bubbles. The two modes therefore differed along a single independent variable: the presence or absence of the virtual companion, as summarized in Table 2.1. The complete trigger message scripts for both modes are provided in Appendix C.

	Mode A (Panel-only)	Mode B (Companion)
Feedback presentation	Panel text prompt	Companion speech bubble
Text style	Mechanical coaching	Conversational
Screenshot		

Table 2.1 Mode A and Mode B comparison

Prior to the main study, a small-scale pilot test was conducted to evaluate the experimental design. The original version of Mode A presented running metrics only and did not include trigger-based feedback messages, whereas Mode B delivered contextual feedback through companion speech bubbles. Pilot testing indicated that this introduced a potential confound, as any observed differences could be attributed either to the presence of the companion or to the presence of encouraging feedback. To ensure that the virtual companion remained the only manipulated factor between conditions, Mode A was revised to include mechanical-toned text prompts using the same trigger logic and thresholds as Mode B. The pilot test also confirmed the stability of BLE connectivity and the reliability of the boundary system during outdoor running.

2.3.2 Procedure

The procedure began with informed consent and a pre-session questionnaire. Participants then completed two 3-minute outdoor running sessions under different experimental

conditions. After each running session, a post-session questionnaire was administered to capture immediate subjective responses. Upon completion of both conditions, participants completed a semi-structured interview. The entire study session lasted approximately 30 minutes.

The study was conducted outdoors on the university sports ground. Due to the Meta Quest 3 boundary system, the running area was restricted to approximately 10-meter by 10-meter. Physical markers were placed at the four corners to help participants identify the boundaries (Figure 2.3).



Figure 2.3 Study venue with bounded running area

2.3.3 Data Collection and Analysis

Subjective data: The pre-session questionnaire collected basic demographics, VR familiarity, pet affinity, and running background. The post-session questionnaire comprised a Companionship scale, a Motivation scale, and a simplified NASA-TLX Workload scale, all on a 1-7 Likert scale. A Simulator Sickness Questionnaire (SSQ) was included in both questionnaires to monitor VR-induced discomfort and was rated on a 0-3 scale. Full questionnaire instruments are provided in Appendix D. Paired-samples Wilcoxon signed-rank tests were used to compare subjective measures across conditions.

Objective data: Heart rate (bpm), pace (min/km), distance (m), and calories (kcal) were logged automatically at one-second intervals throughout each running session. To compare performance differences between conditions, paired-samples t-tests were conducted on the mean values of each measure.

Qualitative data: Semi-structured interviews were conducted after both sessions, exploring five topics: perceived differences between the two modes, memorable reactive moments, perceptions of the companion as a companion versus a visual effect, changes in running rhythm and motivation, and long-term usage intentions. The full interview script is provided in Appendix E. Interviews were audio-recorded with participant consent and transcribed for analysis. The transcripts were analyzed using inductive thematic analysis following Braun and Clarke [12].

Chapter 3

Results

3.1 Pre-Study Surveys

3.1.1 Running Metrics Survey

Around 40 valid responses from participants with regular running habits were included in the analysis. Metrics selected by more than half of respondents were considered to reflect broadly shared user needs while remaining few enough to avoid information overload. As shown in Figure 3.1, five metrics met this threshold: distance covered (74.42%), heart rate (65.12%), elapsed time (62.79%), average speed (53.49%), and calories burned (53.49%). Regarding speed display format, 56% of respondents preferred pace (min/km), 20% preferred speed (km/h), and 24% had no preference (Figure 3.2). As a result, pace was adopted as the display unit.

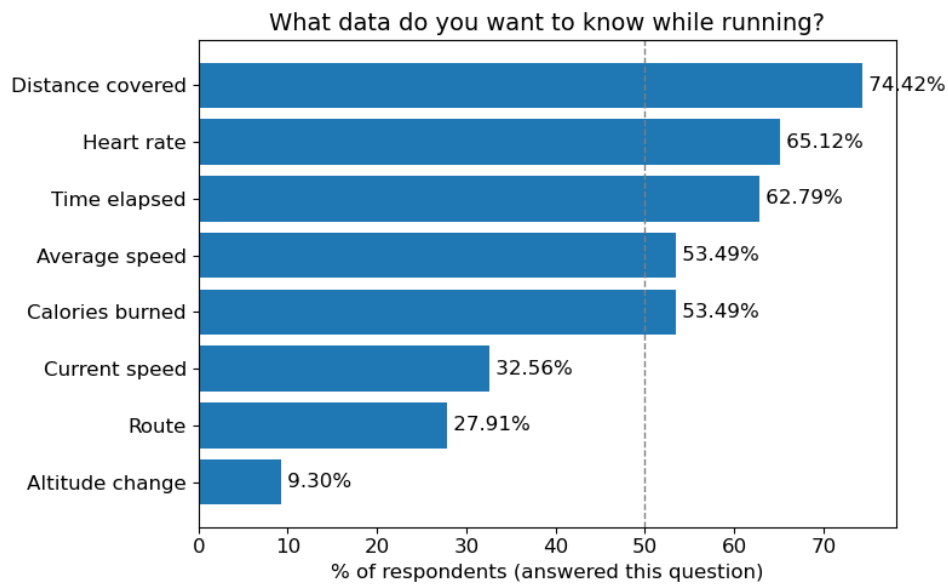


Figure 3.1 Desired running metrics ranked by percentage of respondents

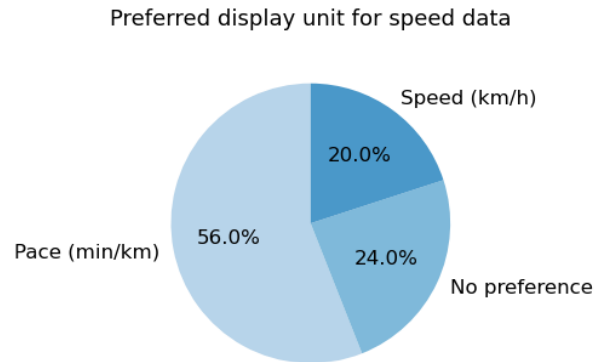


Figure 3.2 Preferred display unit for speed data

3.1.2 Companion Appearance Survey

Approximately 100 responses were collected. As shown in Figure 3.3, the cartoon dog received the highest number of votes overall (26.73%), followed closely by the cartoon rabbit (25.74%). Aggregating results across the three design dimensions independently, as shown in Figure 3.4, participants showed a strong preference for cartoon over realistic renderings (83% vs. 33%) and biological companions over robotic ones (72% vs. 53%). Quadruped locomotion was the most selected movement style (40%). The dominance of cartoon and biological designs is further supported by the free-text reason analysis (Figure 3.5): across both dimensions, the theme of “cuteness” emerged as the most common reason for selection, suggesting that emotional appeal was the primary driver of companion preference.

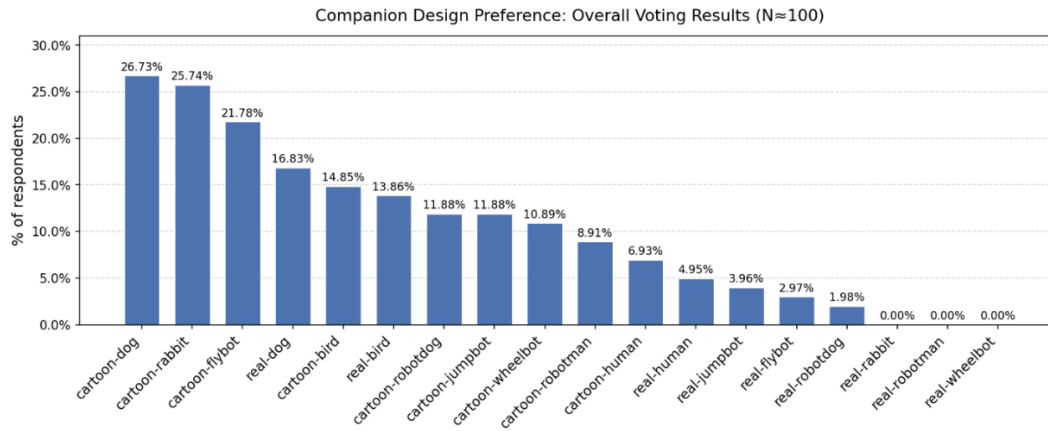


Figure 3.3 Overall companion voting results

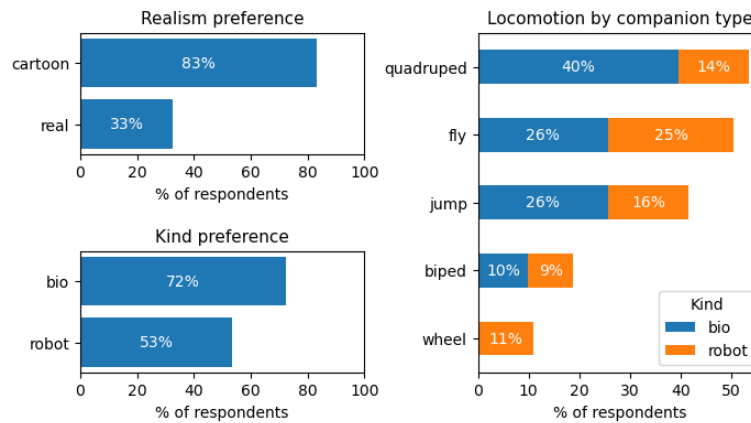


Figure 3.4 Preferences across three design dimensions

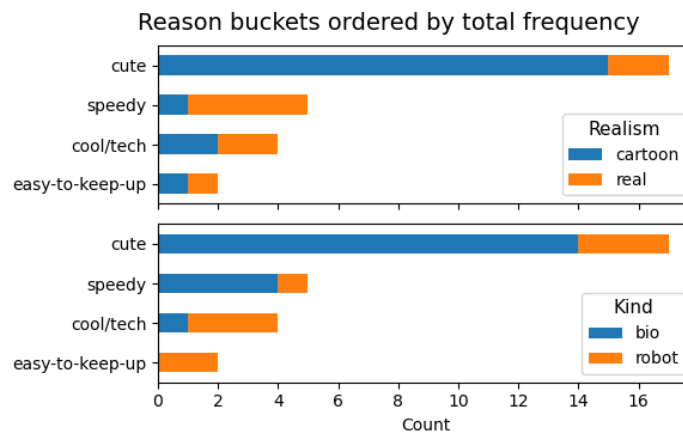


Figure 3.5 Free-text reason themes by realism and kind dimension

3.2 System Implementation

The system guides participants through a series of interface states during each session: device connection, session menu, start screen, running session, and results summary. All interface elements are head-locked, following the participant's position and horizontal rotation to remain consistently within the field of view. All on-screen buttons are operated via the Meta Quest 3's hand-tracking poke interaction. A video demonstration of the complete session flow is available in Appendix F. The following sections describe each interface in detail.

Device Connection: Upon launch, the system scans for and establishes a Bluetooth connection with the wearable heart rate device, displaying a live connection log during this process (Figure 3.6). Once the device is connected and heart rate data is successfully received, the system automatically proceeds to the menu.



Figure 3.6 BLE device connection interface

Menu: The menu screen allows participants to select a participant ID using on-screen plus and minus buttons, and to choose a test mode by tapping either the Mode A or Mode B button (Figure 3.7). Tapping the mode button directly launches the selected session.

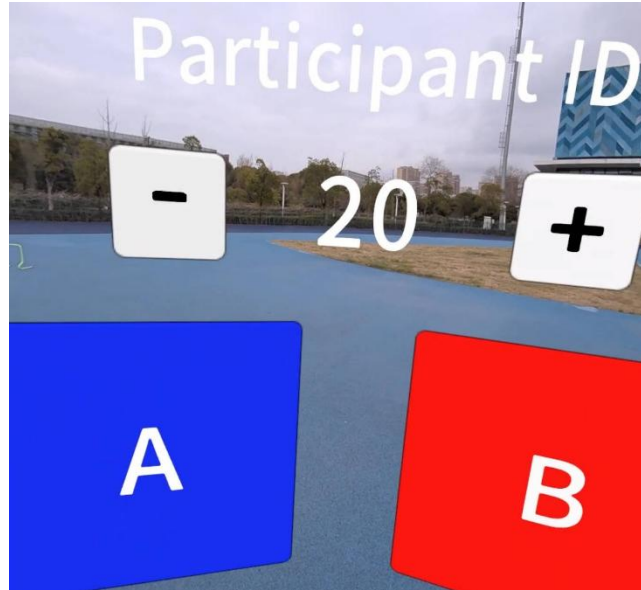


Figure 3.7 Menu interface

Start Screen: After mode selection, a start screen is presented (Figure 3.8). Participants tap the “Start” button when ready to begin the running session, at which point the 3-minute countdown and data recording start simultaneously.



Figure 3.8 Start screen interface

Running Session: During the running session, a HUD is presented to the participant (Figure 3.9). A panel at the top of the display shows elapsed time and distance covered. A panel on the left displays three real-time metrics: heart rate (bpm), average pace (min/km), and calories burned (kcal). Additional feedback is presented on the right side of the display

during the run. In Mode B, this area is occupied by the companion character, with dialogue messages appearing at predefined trigger points. The mode-specific feedback design is described in Section 2.3.1.



Figure 3.9 Running session HUD interface

Results Summary: Upon completion of the 3-minute session, a results panel is displayed showing total time, total distance, average heart rate, average pace, and calories burned (Figure 3.10). A button at the top of the panel returns to the session menu. This placement reduces the risk of accidental activation during the run.



Figure 3.10 Results summary interface

3.3 User study

3.3.1 Participants

Twelve participants (7 female, 5 male, all aged 18-24) were recruited via convenience sampling through social networks and personal contacts. Participants reported a mean VR familiarity score of 3.67/7 and a mean self-rated running ability score of 3.75/7.

3.3.2 Quantitative Results

As shown in Figure 3.11, Mode B produced significantly higher companionship scores than Mode A ($W = 0$, $p = 0.001$). Motivation scores were also significantly higher in Mode B ($W = 0$, $p = 0.001$). In contrast, workload scores showed no significant difference between conditions ($W = 27.5$, $p = 1.0$). These findings suggest that the virtual companion enhanced participants' sense of companionship and motivation without increasing perceived workload. SSQ scores remained low across both conditions, with no notable differences observed between Mode A and Mode B.

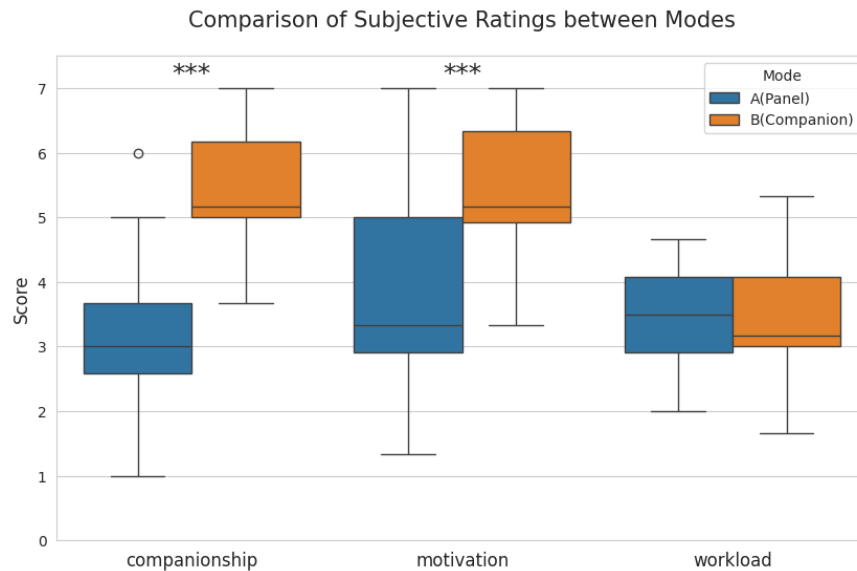


Figure 3.11 Subjective ratings comparison between Mode A and Mode B

3.3.3 Objective Results

Paired t-tests on the four objective physiological metrics found no significant differences between Mode A and Mode B for average heart rate, average pace, distance covered, or calories burned. As shown in Figure 3.12, individual participant trajectories show high variability, but mean values remain largely stable across conditions, confirming that the companion's presence did not systematically alter physical performance.

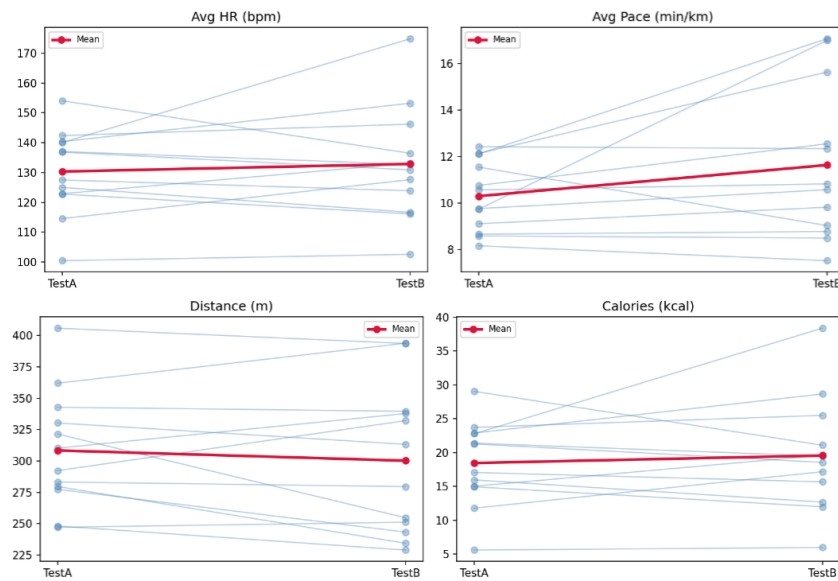


Figure 3.12 Objective physiological metrics across conditions

3.3.4 Qualitative Results

The qualitative analysis revealed three key themes.

First, participants consistently described Mode B as feeling alive rather than merely informative. Many reported that the companion made the running experience more engaging and enjoyable, providing a sense of fun and surprise that was absent from the panel-only condition. As P01 noted, “Having a dog running ahead feels alive — very different from a mechanical panel.” Similarly, P08 described the companion as “full of life, fun, and surprise.”

Second, participants frequently highlighted reactive moments as memorable aspects of the experience. P05 specifically mentioned the dog turning red during periods of elevated heart rate, while P01 recalled the 30-second countdown look-back animation as particularly noticeable. Other commonly cited examples included the heart-rate warning interaction (P04) and the 100-meter milestone celebration (P10). These events were often described as the most engaging and memorable parts of the run.

Third, placement and hardware factors affected usability. Several participants (P03, P06, P11) reported needing to turn their head to view the companion due to its peripheral position, which disrupted the natural running flow. Looking beyond the short-term experience, participants also raised concerns regarding headset weight during running, potential motion sickness in longer sessions, and the social awkwardness of wearing a VR headset in public.

Chapter 4

Discussion, Limitation and Future Work

4.1 Discussion

Overall, the results from quantitative, objective, and qualitative data present a coherent picture: the MR virtual companion enhanced participants' emotional and motivational experience during outdoor running, while having no measurable impact on physiological performance or perceived workload. This convergence across multiple data sources strengthens confidence in the findings.

The increase in companionship and motivation aligns with prior work on virtual companions and relational agents in exercise contexts, where such systems have been shown to enhance engagement and provide sustained motivational support [8,10]. The present findings extend this work to an outdoor MR running setting, demonstrating that similar emotional engagement effects can be achieved in embodied, real-world environments. Importantly, these effects were achieved using a relatively simple companion design without adaptive AI or personalization, suggesting that the threshold for perceived social presence in exercise contexts may be lower than previously assumed. This further indicates that basic reactive behaviors alone may be sufficient to meaningfully enhance the emotional quality of running experiences.

The qualitative findings provide insight into the mechanism underlying these effects. Participants frequently highlighted reactive moments as particularly memorable, especially when the system responded to their physiological state or running progress, such as heart-rate-based color changes and milestone-triggered animations. This indicates that engagement was driven not simply by the presence of a companion, but by its ability to respond dynamically to user state in real time, creating a sense of interaction rather than passive information display.

In contrast, no significant differences were observed in pace control, suggesting that the companion did not influence users' perceived ability to regulate running speed. This likely

reflects the system's design focus on emotional support and feedback rather than pacing guidance. It suggests that emotional engagement and behavioural regulation may require different design mechanisms, and future systems could explore explicit pacing-related companion behaviours.

Similarly, the absence of significant differences in physiological measures indicates that the companion did not impose additional physical or cognitive load during running. Combined with the lack of difference in workload scores, this suggests that the added visual and interactive elements did not interfere with running performance, addressing a key concern for MR systems in dynamic outdoor environments.

Finally, the placement of the companion highlights an unresolved design challenge in outdoor MR systems. Participants reported that the companion's peripheral position required head rotation to maintain awareness, which disrupted the natural running experience. This aligns with prior work on embedded visualization placement in MR running systems [5], and suggests that spatial integration remains a critical open problem for future research.

4.2 Limitation

Several limitations of this study should be acknowledged.

First, the sample size was small ($N = 12$), and all participants were university students within a narrow age range (18-24). This limits the statistical power of the findings and reduces their generalizability to broader populations, including older runners or those with varying fitness levels and VR experience.

Second, each running session lasted only three minutes and was conducted within a bounded 10-meter by 10-meter area on the university sports ground. This setup differs substantially from naturalistic outdoor running, where routes are longer, terrain varies, and runners maintain a consistent direction. The short duration and constrained space may have amplified novelty effects and prevented the emergence of fatigue or motivational patterns that would appear over longer runs.

Third, there was a non-intended design difference between the two modes beyond the independent variable of interest. Mode B included the companion character delivering brief greeting and closing messages on the start and end screens, whereas Mode A presented a minimal interface at these points. Although this difference was not part of the core experimental manipulation, it may have introduced a degree of confounding between conditions.

Fourth, the study employed a single, fixed virtual companion design: a cartoon dog, selected through the pre-study survey. Individual preferences for companion appearance, personality, and interaction style were not accommodated, and it is possible that the companion design suited some participants better than others, contributing to variability in the results.

Fifth, the study was conducted using a Meta Quest 3 headset, which carries practical limitations for outdoor running, including headset weight, limited field of view, and potential social stigma. These hardware constraints likely influenced participants' experiences and may have attenuated the companion's positive effects compared to what could be achieved with lighter or less conspicuous devices in future studies.

4.3 Future Work

Future work could extend the design space of the virtual companion system.

First, companion placement within the visual field remains an open design challenge. Future work should systematically explore head-locked, body-locked, and world-anchored configurations to better understand how spatial positioning affects visibility and the running experience.

Second, future research should explore the integration of personalization and adaptive interaction within companion systems. This includes allowing users to customize the companion's appearance and interaction style, as well as enabling the companion to take a more active role in the running process, for example by running ahead to lead the user to signal an ideal pace. Such developments may further enhance emotional engagement and long-term adoption.

Chapter 5

Conclusion

In conclusion, this project designed, implemented, and evaluated a Mixed Reality virtual running companion for outdoor running. A controlled within-subjects study (N=12) compared the companion condition with a panel-only baseline in a real outdoor setting.

The results showed significant improvements in companionship and motivation, while no apparent differences were found in workload and objective physiological measures, indicating that the companion enhanced the emotional and motivational experience without adding cognitive burden or affecting physical performance. Qualitative findings further highlighted perceived life-likeness, memorable reactive interactions, and placement and usability considerations as key factors shaping user experience.

Overall, the study demonstrates that the emotional dimension of outdoor running, long overlooked by data-centric fitness applications, is addressable through MR companion design. The findings suggest that companionship, rather than performance optimization, may represent a more promising design direction for future running technology, and that even a simple embodied companion can meaningfully alter the affective quality of a solo run.

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Appendix A. Pre-study Surveys

A.1 Running Metrics Preference Survey

跑步数据显示偏好

本问卷旨在了解用户在跑步时的行为习惯偏好，为后续跑步辅助系统的设计提供参考。

您的回答将仅用于学术研究，不会收集任何个人隐私信息。整个问卷约需 1-2 分钟完成。感谢您的参与！

1. 您的性别 / Your gender identification [单选题] *

☐ 男 / Male

☐ 女 / Female

☐ 非二元 / Non-binary

☐ 不愿告知 / Prefer not to say

2. 您的年龄段 / Your age range [单选题] *

☐ 18-24

☐ 25-34

☐ 35-44

☐ 45-54

☐ 55-64

☐ > 65

3. 您平时是否有跑步的习惯？ / Do you have a running habit? [单选题] *

☐ 是 / Yes

☐ 否 / No (Skip to end of survey)

4. 您跑步的频率是？ / How often do you run? [单选题] *

☐ 每天 / On a daily basis

☐ 每周 2-6 次 / 2-6 times per week

☐ 每周 / On a weekly basis

☐ 每月 1-3 次 / 1-3 times per month

☐ 更少 / Less than that

5. 您每次跑步的时长通常是？ / For how long time do you usually run every time? [单选题] *

☐ 5 分钟以内 / Less than 5 min

- 5-15 分钟 / 5-15 min
- 15-30 分钟 / 15-30 min
- 30-60 分钟 / 30-60 min
- 60 分钟以上 / More than 60 min

6. 您在跑步时希望记录或查看哪些数据？（可多选） / What data do you want to know while running? (Multiple choice) [多选题] *

- ☐ 已跑距离 / Distance covered
- ☐ 已跑时长 / Time elapsed
- ☐ 当前速度 / Current speed
- ☐ 平均速度 / Average speed
- ☐ 心率 / Heart rate
- ☐ 消耗卡路里 / Calories burned
- ☐ 路线 / Route
- ☐ 海拔变化 / Altitude change
- ☐ 其他（请注明） / Other (please specify) _____ *

7. 您希望速度数据以哪种方式显示？ / Which display format do you prefer for speed data? [单选题]

* (Displayed only if Q6 includes "Current speed" or "Average speed")

- 配速 / Pace (min/km)
- 时速 / Speed (km/h)
- 没有偏好 / No preference

A.2 Companion Appearance Preference Survey

陪跑精灵形象偏好调查

本问卷旨在了解用户对不同陪跑精灵形象的视觉偏好，以为后续跑步辅助系统的角色设计提供参考。

问卷仅用于学术研究，不会收集任何个人隐私信息，填写约需 1-2 分钟。

若您已填写过“跑步数据显示偏好”问卷，请勿重复填写本问卷。

感谢您的参与！

1. 您的性别 / Your gender identification [单选题] *

- ☐男 / Male
- ☐女 / Female
- ☐非二元 / Non-binary
- ☐不愿告知 / Prefer not to say

2. 您的年龄段 / Your age range [单选题] *

- ☐18-24
- ☐25-34
- ☐35-44
- ☐45-54
- ☐55-64
- ☐ > 65

3. 您平时是否有跑步的习惯？ / Do you have a running habit? [单选题] *

- ☐是 / Yes
- ☐否 / No ([Skip to question 6](#))

4. 您跑步的频率是？ / How often do you run? [单选题] *

- ☐每天 / On a daily basis
- ☐每周 2-6 次 / 2-6 times per week
- ☐每周 / On a weekly basis
- ☐每月 1-3 次 / 1-3 times per month
- ☐更少 / Less than that

5. 您每次跑步的时长通常是？ / For how long time do you usually run every time? [单选题] *

- ☐5 分钟以内 / Less than 5 min
- ☐5-15 分钟 / 5-15 min
- ☐15-30 分钟 / 15-30 min

○30-60 分钟 / 30-60 min

○60 分钟以上 / More than 60 min

6. 下列陪跑精灵形象设计中，您会选择哪一（几）个？ / Among the following running companion designs, which one(s) would you choose? (Select up to 3) [多选题，最多选 3 个] * (The order of the images was randomized for each respondent.)

可在选项下的文字框中填入您选择该形象的原因。 / You may write down your reason for choosing the design in the text box below the option.



□1 _____



□2 _____



□3 _____



□4 _____



□5 _____



□6 _____



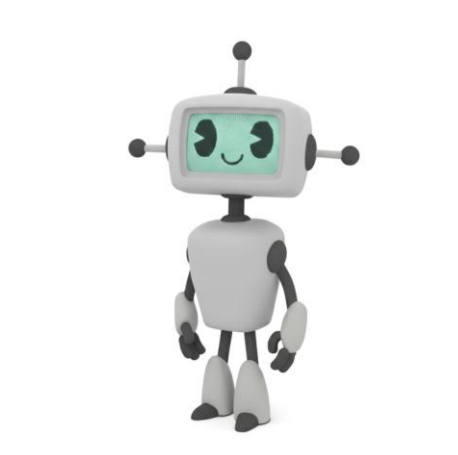
□7 _____



□8 _____



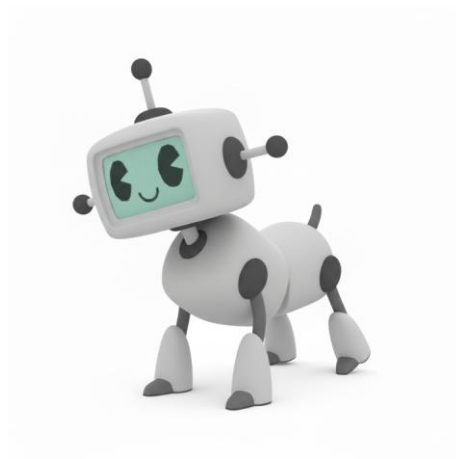
□9 _____



□10 _____



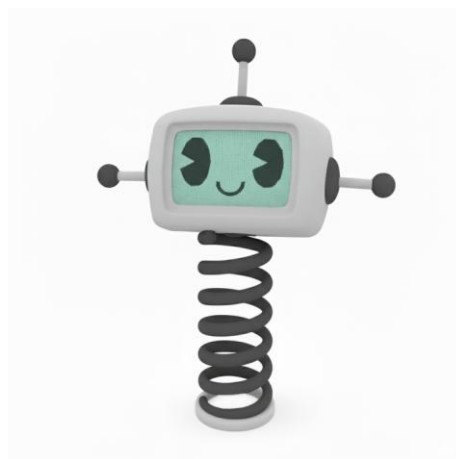
□11 _____



□12 _____



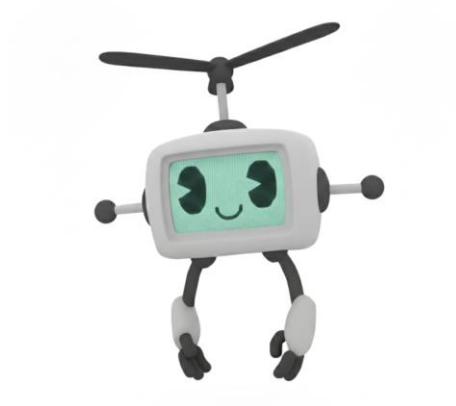
□13 _____



□14 _____



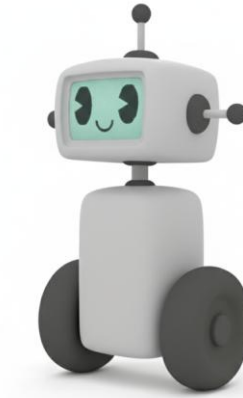
□15 _____



□16 _____



□17 _____



□18 _____

Appendix B. Project Repository

The Unity project for the MR running companion system, including all C# scripts, scene files, and assets, is available at the following repository:

<https://github.com/Donutw/MR-Running-Companion>

Appendix C. Trigger Message Scripts

The following table presents the full set of trigger messages used in Mode A (panel-only) and Mode B (companion) for each trigger event. All messages were delivered in both English and Chinese. The trigger logic and thresholds were identical across both modes; only the tone and phrasing differed.

	A	B
Distance	{0}m reached. Great effort! 已跑{0}米。表现不错！	Amazing! we've gone {0}m together! 太棒了！我们已经一起跑了{0}米啦！
Calories	{0} kcal burned. Well done! 已燃烧{0}千卡。做得好！	Powering up! {0}kcal burned already! 能量全开！已经燃烧了{0}千卡能量！
Heart Rate High	High heart rate. Please slow down. 心跳过快。请适当减速。	Woof! Your heart is beating so fast! Slow down a bit? 汪！你的心跳好快呀！慢一点吧？
Pace Slow	Pace is low. Please speed up a little. 配速偏慢。请适当提速。	Hey! Let's speed up a little! 嘿！我们稍微再加速一点吧！
Stopped	Motion paused. Keep moving! 检测到运动暂停。请继续加油！	Keep going! You can do it! 继续加油！你可以的！
Time left	Last 30 seconds! Finish strong! 最后 30 秒！坚持到底！	Last 30 seconds! Let's finish this together! 最后 30 秒啦！一起坚持到底吧！

Appendix D. User Study Questionnaires

D.1 Pre-Test Questionnaire

VR 户外跑步陪伴系统 / Evaluation of VR Companion in Outdoor Running

前置问卷 / Pre-Study Questionnaire

您好！感谢您参与本次关于 VR 户外跑步陪伴系统的研究。Hello! Thank you for participating in this research on the VR outdoor running companion system.

本研究旨在评估 VR 环境中不同视觉反馈模式对用户运动体验、陪伴感及感知费力程度的影响。本次实验总时长约为 30 分钟，包含前置问卷、两次 3 分钟的户外慢跑测试及后置问卷/访谈。This study aims to evaluate the impact of different visual feedback modes in a VR environment on user experience, sense of companionship, and perceived exertion. The total duration of this session is approximately 30 minutes, including a pre-study questionnaire, two 3-minute outdoor jogging sessions, and post-study questionnaires/interviews.

隐私与匿名声明 / Privacy and Anonymity:

- 您所提供的的数据均将以匿名方式处理，仅用于学术研究。
- 原始数据将存储在受密码保护的设备中，不会泄露您的个人信息。

- 参与本次实验完全出于自愿，您可以随时要求中止实验而无需提供任何理由。
- All data provided will be processed anonymously and used for academic research purposes only.
- Raw data will be stored on password-protected devices, and no personally identifiable information will be disclosed.
- Participation is voluntary, and you are free to withdraw from the study at any time without giving a reason.

安全提示 / Safety Notice: 实验过程中请务必听从实验员的引导，并在指定的安全区域内活动。如在佩戴 VR 头显过程中感到任何眩晕或不适，请立即停止实验并告知实验员。Please follow the guidance of the researcher and remain within the designated safe area during the experiment. If you experience any dizziness or discomfort while wearing the VR headset, please stop immediately and inform the researcher.

一、基础信息 / Basic Information

1. 受测者编号 / Participant ID: [填空题] *

2. 您的性别 / Your gender identification: [单选题] *

- ☐ 男 / Male
- ☐ 女 / Female
- ☐ 非二元 / Non-binary
- ☐ 不愿告知 / Prefer not to say

3. 您的年龄段 / Your age range: [单选题] *

- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45-54
- ☐ 55-64
- ☐ > 65

4. 您对 VR（虚拟现实）设备的熟悉程度 / How familiar are you with VR devices? [单选题] *

1 = 完全

不熟悉 /

Not at

all

☐1

☐2

☐3

☐4

☐5

☐6

☐7

7 = 非常

熟悉 /

Extremely

ly

familiar

5. 您在现实生活中对宠物的喜爱程度？ / How much do you like pets in real life? [单选题] *

1 = 完全

不喜欢 /

Not at

all

☐1

☐2

☐3

☐4

☐5

☐6

☐7

7 = 非常

喜欢 /

Extremely

ly

二、跑步习惯与态度 / Running Background & Attitudes

6. 您的跑步频率 / How often do you run? [单选题] *

☐每天 / Daily

☐每周 2-6 次 / 2-6 times per week

☐每周 / On a weekly basis

☐每月 1-3 次 / 1-3 times per month

☐更少 / Less than that

7. 您对自己的跑步能力评价是？ / How would you rate your running ability? [单选题] *

1 = 极低

/ Very

low

☐1

☐2

☐3

☐4

☐5

☐6

☐7

7 = 极高

/ Very

high

8. 您是否享受跑步的过程？ / Do you enjoy the process of running? [单选题] *

1 = 完全

不享受 /

Not at

all

☐1

☐2

☐3

☐4

☐5

☐6

☐7

7 = 非常

享受 /

Extremely

ly

9. 您是否觉得独自跑步很枯燥？ / Do you find running alone boring? [单选题] *

1 = 一点								7 = 非常
也不枯								枯燥 /
燥 / Not	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	Extreme
at all								ly

10. 在跑步时，您是否渴望有虚拟伙伴陪同？ / Do you desire a virtual companion while running?

[单选题] *

1 = 完全								7 = 非常
不渴望 /								渴望 /
Not at	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐ 6	☐ 7	Extreme
all								ly

三、当前身体状态评估 / Simulator Sickness Questionnaire (SSQ)

请根据您当前的感受，对以下症状的严重程度进行打分. Please rate the severity of each symptom based on your current state.

11. 头部是否有疼痛、沉重或压迫感？ / Do you have any headache, heaviness, or pressure in your head? [单选题] *

0 = 无 / None	☐ 0	☐ 1	☐ 2	☐ 3	3 = 严重 /
					Severe

12. 眼睛是否感到酸痛、干涩或疲劳？ / Do your eyes feel sore, dry, or tired? [单选题] *

0 = 无 / None	☐ 0	☐ 1	☐ 2	☐ 3	3 = 严重 /
					Severe

13. 是否有反胃、胃部不适或想呕吐的感觉？ / Do you feel nauseous or have stomach discomfort?

[单选题] *

0 = 无 / None	☐ 0	☐ 1	☐ 2	☐ 3	3 = 严重 /
					Severe

14. 是否感到身体疲惫、无力或精力耗尽？ / Do you feel physically tired, weak, or exhausted? [单选题] *

0 = 无 / None ☐0 ☐1 ☐2 ☐3 3 = 严重 / Severe

15. 是否有头晕、天旋地转或站不稳的感觉？ / Do you feel dizzy or lightheaded? [单选题] *

0 = 无 / None ☐0 ☐1 ☐2 ☐3 3 = 严重 / Severe

16. 是否感到身体动作与视觉画面之间不一致/不协调？ / Do you feel a lack of coordination between your body movements and what you see? [单选题] *

0 = 无 / None ☐0 ☐1 ☐2 ☐3 3 = 严重 / Severe

D.2 Post-Test Questionnaire

VR 户外跑步陪伴系统 / Evaluation of VR Companion in Outdoor Running

实验后置问卷 / Post-Session Questionnaire

您好！感谢您完成本阶段的跑步测试。Hello! Thank you for completing this session of the running test.

这份问卷旨在收集您**刚刚完成的这段跑步体验**中的即时感受。请在填写时仅针对刚才测试的模式（模式 A 或 模式 B）进行评价。本问卷预计耗时 3-5 分钟。This questionnaire is designed to collect your **immediate feelings** about the running experience you have **just completed**. When answering, please focus solely on the mode you just tested (Condition A or Condition B). This will take approximately 3-5 minutes.

填写须知 / Instructions:

- - 请务必确认您选择的模式（A 或 B）与实验员分配给您的测试顺序一致。
- - 请根据您此刻最真实的身体状况和心理感受作答。
- - 您的数据将通过受测者编号与前置问卷进行匹配，所有信息将严格保密，仅用于学术研究。
- - Please ensure that the mode you select (Condition A or B) matches the sequence assigned by the researcher.

- - Please answer based on your most genuine physical and psychological state at this moment.
- - Your data will be matched with your pre-study questionnaire using your Participant ID. All information will remain strictly confidential and be used for academic research only .

再次感谢您的配合。如有任何不适，请及时告知实验员。 Thank you again for your cooperation. If you feel any discomfort, please inform the researcher immediately.

一、识别信息 / Identification

1. 受测者编号 / Participant ID: [填空题] *

2. 请选择您刚才完成的测试模式 / Which mode did you just complete? [单选题] *

☐ A

☐ B

二、陪伴感评估 / Companionship (Social Presence)

3. 我感到自己不是一个人在跑步。 / I felt like I was not running alone. [单选题] *

1= 完全

7 = 非常

不同意 /

同 意 /

Not at

Extreme

all

ly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

4. 我能感觉到某种“生命力”在陪着我。 / I could sense a certain "life-likeness" accompanying me.

[单选题] *

1= 完全

7 = 非常

不同意 /

同 意 /

Not at

Extreme

all

ly

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

5. 刚才的界面让我觉得运动过程不孤单。 / The interface made me feel that I was not alone during

the exercise. [单选题] *

1= 完全

7 = 非常

不同意 /

同 意 /

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

Extremely

6. 我觉得刚才这种跑步方式非常有趣。 / I found this way of running very enjoyable. [单选题] *

7 = 非常
同意 /
Extreme
ly

1= 完全不同意 / Not at all 7= 非常同意 / Extremely

1= 完全不同意 / Not at all 7= 非常同意 / Extremely

9. 读取和理解数据需要花费您很多精力吗？ / How much mental effort was required to read and understand the data? [单选题] *

7 = 极高
/ Very
High

39

1= 极低 7 = 极高
 / Very ☐1 ☐2 ☐3 ☐4 ☐5 ☐6 ☐7 / Very
 Low High

11. 您觉得为了完成这段跑步，您付出了多少努力？ / How hard did you have to work to accomplish this run? [单选题] *

1= 极低 7 = 极高
 / Very ☐1 ☐2 ☐3 ☐4 ☐5 ☐6 ☐7 / Very
 Low High

五、当前身体状态评估 / Simulator Sickness Questionnaire (SSQ)

请根据您当前的感受，对以下症状的严重程度进行打分. Please rate the severity of each symptom based on your current state.

12. 您的头部是否有疼痛、沉重或压迫感？ / Do you have any headache, heaviness, or pressure in your head? [单选题] *

0 = 无 / None ☐0 ☐1 ☐2 ☐3 3 = 严重 / Severe

13. 您的眼睛是否感到酸痛、干涩或疲劳？ / Do your eyes feel sore, dry, or tired? [单选题] *

0 = 无 / None ☐0 ☐1 ☐2 ☐3 3 = 严重 / Severe

14. 您是否有反胃、胃部不适或想呕吐的感觉？ / Do you feel nauseous or have stomach discomfort? [单选题] *

0 = 无 / None ☐0 ☐1 ☐2 ☐3 3 = 严重 / Severe

15. 您是否感到身体疲惫、无力或精力耗尽？ / Do you feel physically tired, weak, or exhausted? [单选题] *

0 = 无 / None ☐0

☐1

☐2

☐3

3 = 严重 /
Severe

16. 您是否有头晕、天旋地转或站不稳的感觉？ / Do you feel dizzy or lightheaded? [单选题] *

0 = 无 / None ☐0

☐1

☐2

☐3

3 = 严重 /
Severe

17. 您是否感到身体动作与视觉画面之间不一致/不协调？ / Do you feel a lack of coordination between your body movements and what you see? [单选题] *

0 = 无 / None ☐0

☐1

☐2

☐3

3 = 严重 /
Severe

Appendix E. Semi-Structured Interview Script

Semi-Structured Interview Script / 半结构化访谈提纲

Instructions: The following questions will be asked in a semi-structured interview format after the participant has completed both running sessions. The interview will be audio-recorded for later analysis.

说明：以下问题将在受试者完成两次跑步测试后，以半结构化访谈的形式进行提问。访谈过程将进行录音以便后续分析。

1. What was the biggest difference you felt between these two running modes (Condition A: Data-only vs. Condition B: Virtual Companion)?

这两种跑步方式（模式 A：纯数据 vs 模式 B：精灵陪跑）给您的感觉最大的不同是什么？

2. Was there a specific moment (e.g., the dog changing color or a bubble popping up) where you clearly noticed the companion responding to your running status?

“在跑步过程中，有没有哪个具体的时刻（例如 A 模式的文字弹出或 B 模式的小狗变化），让您明显感觉到系统在回应您的跑步状态？”

3. Did the virtual dog feel more like a "companion" to you, or was it more like a "visual effect" or a "distraction"? Why?

您觉得小狗的存在更像是一种“陪伴”，还是更像一种“视觉特效”或“干扰”？为什么？

4. During the actual running session, did the virtual companion lead to any changes in your rhythm or motivation?

在实际跑步过程中，陪跑精灵有没有让您在节奏或动力上做出任何改变？

5. Ignoring hardware limitations (e.g., headset weight), would you consider using this "virtual companion" for your long-term outdoor running? What potential issues do you foresee?

如果不考虑硬件限制（如头显重量），您觉得自己会长期使用这个“陪跑精灵”进行户外跑步吗？
您觉得长期使用可能会遇到什么问题？

Appendix F. Demo Video

A demonstration video of the MR running companion system in operation, including both Mode A and Mode B, is available at the following link:

Vimeo: <https://vimeo.com/1174433829>

Bilibili: <https://www.bilibili.com/video/BV1B4wRzgE7b>